Special instructions for students

Unless otherwise stated the following parameters have the following definitions:

V_{DS} *: Drain Source Voltage.*

V_{GS} *: Gate Source Voltage.*

V_{TH} *: Threshold Voltage.*

U_t *: Thermal Voltage.*

n : Weak Inversion Slope factor.

g_m*: Transconductance.*

Solutions

1.

a) Describe in your own words what the benefits are of using current mode rather than voltage mode.

The student is expected to expand a discussion highlighting the following points as a benefit for current mode:

- **Higher bandwidth** – no slew rate limitation since current is input
- **Higher Dynamic Range** – Max signal does not depend entirely on VDD
- **Work with smaller supplies** – Good for low power
- **Easier Computation** through translinear, KCL

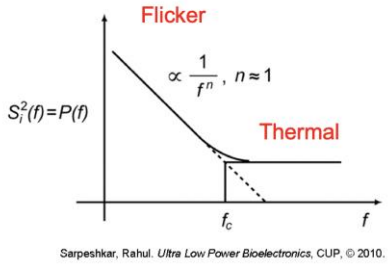
[5]

b) Analogue circuits are better for signal processing than digital circuits. Prove this statement from first principle MOSFET noise equations and discuss in your own words for which scenarios this is true.

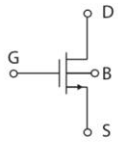
Starting from first principles the student is expected to derive the noise resource equation for analogue circuits and then describe for which precision this is suitable for signal processing.

Analogue is noise limited

MOS transistor dominated by Thermal and Flicker Noise



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$$\overline{v_n^2} = \int_{f_l}^{f_h} \left(\frac{8KT}{3gm}\right) \cdot df + \left(\frac{k_f}{C_{ox} \cdot W \cdot L}\right) \frac{df}{f}$$

Thermal Flicker

Where

K=	Boltzmann's constant (J/K)
T=	Temperature (K)
g_m=	Transconductance (S)
k_f=	Flicker noise coefficient (V^2F)
C_{ox}=	Unit oxide capacitance (F/m^2)
W=	Transistor width (m)
L=	Transistor length (m)

Consider a MOS transistor

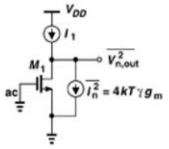
Thermal Noise due to drain resistance: $I_n^2 = 4KT(\frac{2}{3}gm) \cdot \Delta f$

Referring Noise sources to the input: $v_n^2 = \frac{I_n^2}{gm^2} = \frac{8}{3}KT\frac{\Delta f}{gm}$

Substituting for current: Strong Inversion - $g_m = 2\sqrt{\beta I_D} = I_D^{0.5} \cdot \sqrt{4\beta}$

Weak Inversion - $g_m = I_D^1 / nU_t$

Input referred thermal noise equation: Where P=1 for WI and P=0.5 for SI $v_n^2 = \frac{K_W(p)}{I_D^p} \cdot \Delta f$ **Current dependent !**



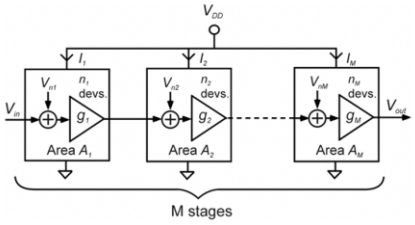
Flicker Noise: $v_n^2 = \frac{k_f}{C_{ox}WL} \cdot \ln\left(\frac{f_h}{f_l}\right) = \frac{K_f}{A} \cdot \ln\left(\frac{f_h}{f_l}\right)$ **Area dependent ! *A=W.L***

Total MOS Noise: $v_n^2 = \frac{K_w}{I_D^p} \cdot \Delta f + \frac{K_f}{A} \cdot \ln\left(\frac{f_h}{f_l}\right)$ Bandwidth $\Delta f = f_h - f_l$

Cascading analogue stages

- Early computational stages contribute more noise than late stages
- Must be kept to a minimum

MOS noise: $$v_n^2 = \frac{K_w}{I_D^p}\cdot\Delta f + \frac{K_f}{A}\cdot\ln\left(\frac{f_h}{f_l}\right)$$



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Example

Assume a cascade of ASP blocks, with n_i devices, a gain g_i, Total area A_i, and additive noise V_{n1}:

Distributed Gain : $$G_i = \prod_{k=i}^{k=M} g_k$$

Total Output Noise: $$v_{no}^2 = \sum_{i=1}^{i=M} v_{ni}^2 G_i^2$$

$$v_{no}^2 = \sum_{i=1}^{i=M} v_{ni}^2 G_i^2 = \sum_{i=1}^{i=M}\left(n_i \frac{K_w}{(I_i/n_i)^p}\cdot\Delta f + n_i\frac{K_f}{(A_i/n_i)}\cdot\ln\left(\frac{f_h}{f_l}\right)\right)G_i^2$$

Where n_i is the number of devices

Noise resource equation

What does the the noise resource equation tell us:

$$v_{no}^2 = \sum_{i=1}^{i=M}\left(n_i \frac{K_w}{(I_i/n_i)^p}\cdot\Delta f + n_i\frac{K_f}{(A_i/n_i)}\cdot\ln\left(\frac{f_h}{f_l}\right)\right)G_i^2$$

- Increasing Power and Area reduce the noise
- Noise is accumulative with the number of stages
- If we are willing to compute slowly (Δf and f_h/f_l are both small) then we can keep output noise low without consuming a lot of power.
- If noise is to be minimised more resources should be distributed to the parts of a system that affect all other parts(i.e initial stages) and those which are complex (high n_i).

More generalised form for whole system relating power and area:

$$v_{no}^2 = \frac{C_w}{P_T^p}\Delta f + \frac{C_f}{A_T}\ln\left(\frac{f_h}{f_l}\right)$$

where: $$P_T = V_{DD} I_T \qquad \sum_i I_i = I_T \qquad \sum_i A_i = A_T$$

SOLUTIONS

The costs of SNR in analog systems

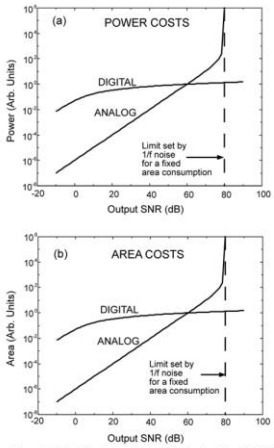
Noise Resource Equation: $v_{no}^2 = \frac{C_w}{P_T^p}\Delta f + \frac{C_f}{A_T}\ln\left(\frac{f_h}{f_l}\right)$

Max output SNR: $SNR = \frac{V_{DD}^2}{v_{no}^2}$

Power vs SNR: $P_T = \left(\frac{C_w \cdot SNR \cdot \Delta f}{V_{DD}^2 - SNR \cdot (C_f \ln(f_h / f_l)/A_T)}\right)^{1/p}$

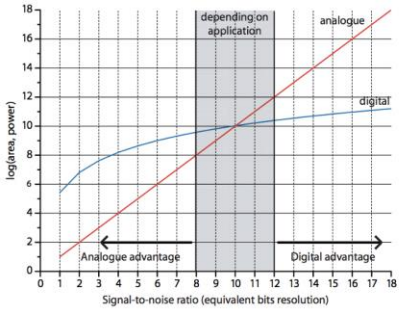
Area vs SNR: $A_T = \left(\frac{C_f \cdot SNR \cdot \ln(f_h / f_l)}{V_{DD}^2 - SNR \cdot (C_w \cdot \Delta f / P_T^p)}\right)$

Resource Precision Equations



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The analogue/digital Trade off



Number of bits:

$$\mathrm{DR_{dB}} = \mathrm{SNR_{dB}} = 20\log_{10}(2^n) \approx 6.02 \cdot n$$

- Analogue has clear advantage when we can afford to be less precise (<8 bits)

[10]

c) For a given mixed mode sensing system, analogue signals need to be digitized for data transmission. What is the best strategy to implement such as system which minimizes power consumption? You are required to describe this in your own words and show an example of an optimum system.

The student is expected to expand a discussion including the following points.

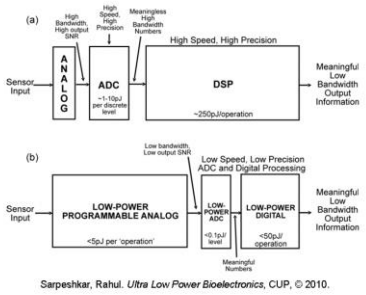
The optimum point for digitisation

- Increasing the Analogue processing can decrease the power of the system as a whole!
- By consuming a bit more power in the analogue front end we can reduce the digital power drastically!

Example: Mobile Phones

Data Input: 1Ghz
16 Bit Direct Quantisation: 262W!
(assuming 1pJ-per-quantisation level and 2*Nyquist sampling)
Power=$4x10^{9}*2^{16}x10^{-12}$=262W
Processing power=4 Gsamples/s

Can we do better with some ASP?



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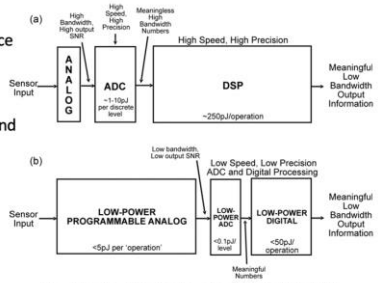
The optimum point for digitisation

Better Strategy:

- Use the analogue for intelligent processing to reduce the bandwidth to the ADC converter.
- Use Analogue Mixers to convert high Frequency signals (1Ghz) to audio base band (20Khz). Then digitize at much lower frequencies to save power and computation!

Audio: 20Khz
Power Cost of Analogue: 10mW
Total Power= 10mW+$4*20^{3}*2^{16}x10^{-12}$=15mW!
Processing power: 80 Ksamples/s

- 3 Orders of Magnitude saving in power!
- More relaxed processing requirements.



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[5]

Solutions

2. This question concerns translinear Circuits.

a) State and derive the translinear principle (TLP) for a loop of MOS transistors working in weak inversion and discuss the limitations. Give all assumptions you make.

[bookwork]

In a closed loop containing an even number of forward biased junctions, arranged so that there are an equal number of clockwise facing and counterclockwise facing polarities, the product of the current densities in the clockwise direction is equal to the product of the current densities in the counterclockwise direction.

[2 points for definition]

Number of Clockwise Junctions (Voltage Rises)=Number of Anticlockwise Junctions (Voltage drops)

$$\sum_{n\in CW} V_n = \sum_{n\in CCW} V_n$$

$$V_n = nU_T \ln\left(\frac{I_n}{I_{DO}}\right)$$

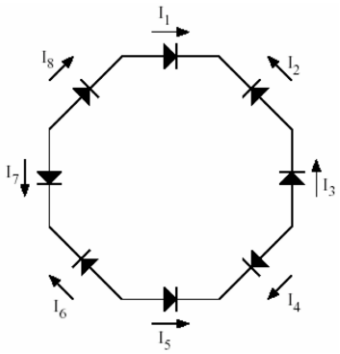
$$\sum_{n\in CW} nU_T \ln\left(\frac{I_n}{I_{DO}}\right) = \sum_{n\in CCW} nU_T \ln\left(\frac{I_n}{I_{DO}}\right)$$

$$\sum_{n\in CW} \ln\left(\frac{I_n}{I_{DO}}\right) = \sum_{n\in CCW} \ln\left(\frac{I_n}{I_{DO}}\right)$$

$$\prod_{n\in CW} \frac{I_n}{I_{DO}} = \prod_{n\in CCW} \frac{I_n}{I_{DO}}$$

$$\prod_{n\in CW} \frac{I_n}{I_n} = I_{DO}^{CW-CCW} \prod_{n\in CCW} \frac{I_n}{I_n}$$

$$\prod_{n\in CW} \frac{I_n}{I_n} = \prod_{n\in CCW} \frac{I_n}{I_n}$$



Assumptions: Temperature and saturation currents are all the same.

[3 points for derivation]

[5]

b) For the circuit shown Figure 2.1, I_{X1} and I_{X2} are input currents whereas I_Z is the output current. Express the current I_Z as a function of the currents I_{X1} and I_{X2} and the transistor area W. When $I_{X1}=I_{X2}=I$, determine the value of the transistor area K for which $I_Z=I$.

[10]

Following translinear loop 1,

$$\left(\frac{I_{x1}}{2}\right)\times\left(\frac{I_{x1}}{2}\right)\times\left(\frac{I_{x1}}{2}\right)\times I_0 = I_0 \times \left(\frac{I_{z1}}{K}\right)\times I_z \times I_z$$

$$\frac{I_{x1}^3}{8} = \frac{I_{z1} I_z^2}{K}$$

$$I_{z1} = \frac{K}{8}\frac{I_{x1}^3}{I_z^2} \quad (1)$$

Following translinear loop 2,

Solutions

$$\left(\frac{I_{x2}}{2}\right)\times\left(\frac{I_{x2}}{2}\right)\times\left(\frac{I_{x2}}{2}\right)\times I_0 = I_0\times\left(\frac{I_{z2}}{K}\right)\times I_z\times I_z$$

$$\frac{I_{x2}^3}{8} = \frac{I_{z2}I_z^2}{K}$$

$$I_{z2} = \frac{K}{8}\frac{I_{x2}^3}{I_z^2} \quad (2)$$

Where

$$I_z = I_{z1} + I_{z2} \quad (3)$$

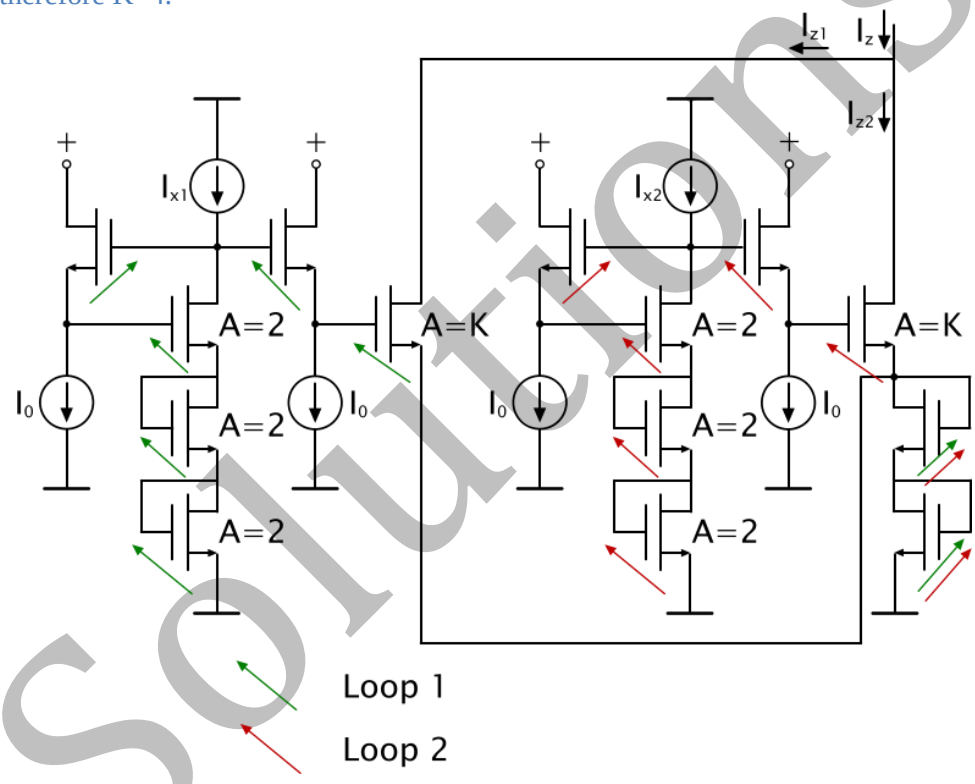
Hence

$$I_z = \frac{K}{8}\frac{I_{x1}^3 + I_{x2}^3}{I_z^2}$$

The transfer function is given by

$$I_z^3 = \frac{K}{8}(I_{x1}^3 + I_{x2}^3)$$

And therefore K=4.



c) Figure 2.2 shows a translinear analogue multiplier. Derive its transfer function and show how this circuit can be modified to improve its gain. Derive the differential gain for a two stage multiplier consisting of the improved circuit.

Translinear circuit yields:

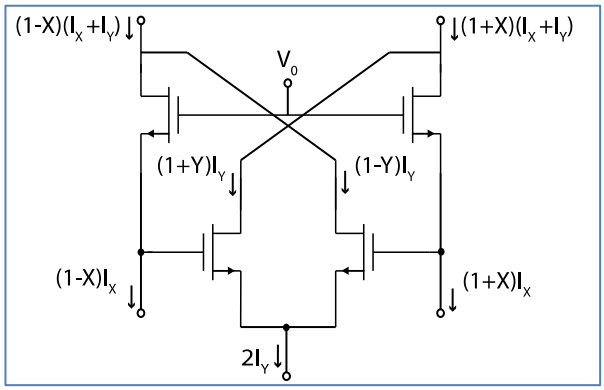
$$(1-X)I_X(1+Y)I_Y = (1+X)I_X(1-Y)I_Y$$

And with X=Y

$$\frac{I_{out}}{I_{in}} = \frac{(1+Y)I_Y - (1-Y)I_Y}{(1+X)I_X - (1-X)I_X} = \frac{2YI_Y}{2XI_X} = \frac{I_Y}{I_X}$$

Modification to gilbert gain Cell to improve gain:

Solutions



Transfer function:

$$\frac{I_{out}}{I_{in}} = \frac{2X(I_X + I_Y)}{2XI_X}$$

For two stages:

$$\frac{I_{out}}{I_{in}} = \frac{(I_X + I_Y)}{I_X}\frac{(I_X + I_Y)}{I_X} = \frac{(I_X + I_Y)^2}{I_X^2}$$

[5]

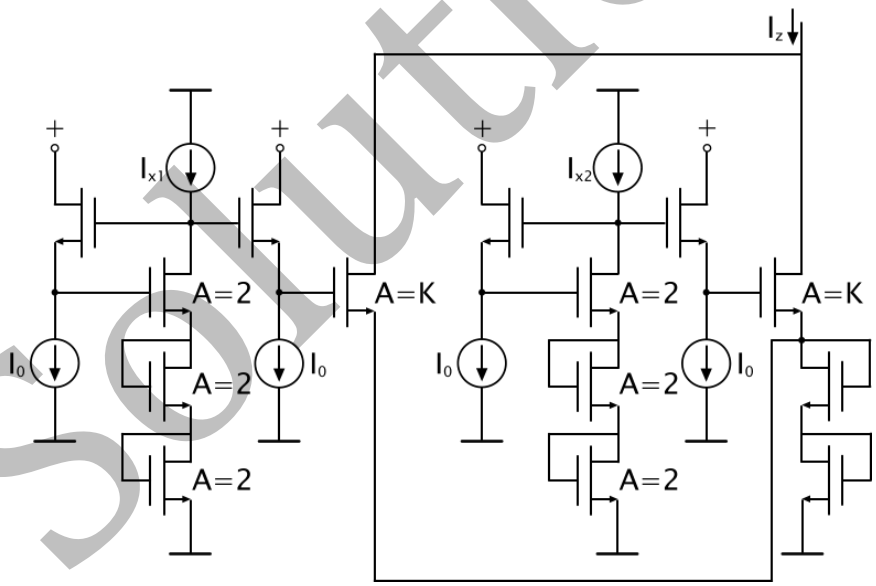


Figure 2.1

Solutions

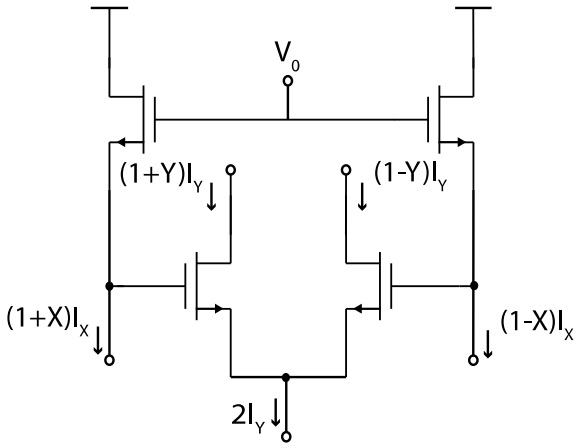


Figure 2.2

3.

a) Figure 3.1 shows a block diagram of a current-mode companding integrator, with an input X and an output Y. f() and g() represent non-linear functions which compress and expand the signal such that the input-output relationship of the integrator is linear and given by:

$$Y = t \int X \times dt$$

Given that the expansive function is defined as $I_{out} = g(V_c) = I_T \tanh(V_c / nU_t)$, derive the function f(x) such that linear integration is achieved. All constants should be grouped to represent a current I_1, with the function f() composed of just I_{in}, I_1 and I_T.

New derivation

Requirement for linear integration: $Y = \tau \int X \cdot dt$ i.e $\frac{dY}{dt} = \tau X$

$$\frac{dY}{dt} = \frac{d(g(V_c))}{dV_c} \cdot \frac{dV_c}{dt} = \frac{d(g(V_c))}{dV_c} \cdot \frac{I_c}{C} = \frac{d(g(V_c))}{dV_c} \cdot \frac{f(X)}{C}$$

$\frac{dY}{dt} = \tau X$ thus $\frac{d(g(V_c))}{dV_c} \cdot \frac{f(X)}{C} = \tau X$ so for **linear integration:** $f(X) = \tau X C \left(\frac{d(g(V_c))}{dV_c} \right)^{-1}$

$$\frac{d(g(V_c)}{dV_c} = \frac{I_T}{nU_t} \sec^2\left(\frac{V_c}{nU_t}\right) = \frac{I_T}{nU_t}\left(1 - \tan^2\left(\frac{V_c}{nU_t}\right)\right)$$

$$\tanh\left(\frac{V_c}{nU_t}\right) = I_{out}/I_T$$

$$\frac{d(g(V_c)}{dV_c} = \frac{I_T}{nU_t}\left(1 - \left(\frac{I_{out}}{I_T}\right)^2\right)$$

$$f(x) = \frac{I_{in}\tau C n U_t}{I_T - I_{out}^2/I_T} = I_{in}\frac{I_1 I_T}{I_T^2 - I_{out}^2}$$

[6]

b) The transfer function of a second order topology has been decomposed into the following state-space equations:

Solutions

$$\dot{X_1} = -\left(\frac{\omega_0}{2Q}\right)X_1 - \omega_0\left(1 - \frac{1}{4Q^2}\right)X_2 + \omega_0 U$$
$$\dot{X_2} = -\left(\frac{\omega_0}{2Q}\right)X_2 + \omega_0 X_1$$
$$Y_1 = X_1$$
$$Y_2 = X_2$$

whereby Y_1 and Y_2 are the outputs, X_1 and X_2 are the state-variables and U is the input.

i. Show that the output Y_1 can be used to implement a second order lowpass transfer function and the output Y_2 can be used to implement a "two-pole one-zero" second order transfer function.

New derivation:

$$\left[s + \frac{\omega_0}{2Q}\right]X_1(s) - \omega_0\left(1 - \frac{1}{4Q^2}\right)X_2(s) = \omega_0 U$$
$$-\omega_0 X_1(s) + \left[s + \frac{\omega_0}{2Q}\right]X_2(s) = \omega_0 U(s)$$

Substituting for $X_2(s)$ (2 points)

$$\frac{Y_1(s)}{U(s)} = \frac{X_1(s)}{U(s)} = \frac{\omega_0^2}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2}, \textit{low pass response}$$

Substituting for $X_1(s)$ (2 points)

$$\frac{Y_2(s)}{U(s)} = \frac{X_2(s)}{U(s)} = \frac{\omega_0\left(s + \frac{\omega_0}{2Q}\right)}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2}, \textit{two pole , one zero response}$$

[4]

ii. By using the mappings below, show how these state space equations can be mapped to non-linear log-domain design equations. State any assumptions you make.

$$X_1 = I_1 \exp\left(\frac{V_1}{nU_t}\right) \qquad X_2 = I_2 \exp\left(\frac{V_2}{nU_t}\right) \qquad U = I_U \exp\left(\frac{V_U}{nU_t}\right)$$

New derivation from Theory:

$$\frac{I_1}{nU_t}\exp\left(\frac{V_1}{nU_t}\right)\dot{V_1} = -\left(\frac{\omega_0}{2Q}\right)I_1 \exp\left(\frac{V_1}{nU_t}\right) - \omega_0\left(1 - \frac{1}{4Q^2}\right)I_2 \exp\left(\frac{V_2}{nU_t}\right) + \omega_0 I_u \exp\left(\frac{V_u}{nU_t}\right)$$
$$\frac{I_2}{nU_t}\exp\left(\frac{V_2}{nU_t}\right)\dot{V_2} = -\left(\frac{\omega_0}{2Q}\right)I_2 \exp\left(\frac{V_2}{nU_t}\right) + \omega_0 I_1 \exp\left(\frac{V_1}{nU_t}\right)$$

$$C\dot{V_1} = -CnU_t\left(\frac{\omega_0}{2Q}\right) - CnU_t\omega_0\left(1 - \frac{1}{4Q^2}\right)\frac{I_2}{I_1}\exp\left(\frac{V_2 - V_1}{nU_t}\right) + CnU_t\omega_0\frac{I_u}{I_1}\exp\left(\frac{V_u - V_1}{nU_t}\right)$$
$$C\dot{V_2} = -CnU_t\left(\frac{\omega_0}{2Q}\right) + CnU_t\omega_0\frac{I_1}{I_2}\exp\left(\frac{V_1 - V_2}{nU_t}\right)$$

Let C1=C2=C and define: $I_\omega = CnU_t\omega_0$

A suitable mapping:

Solutions

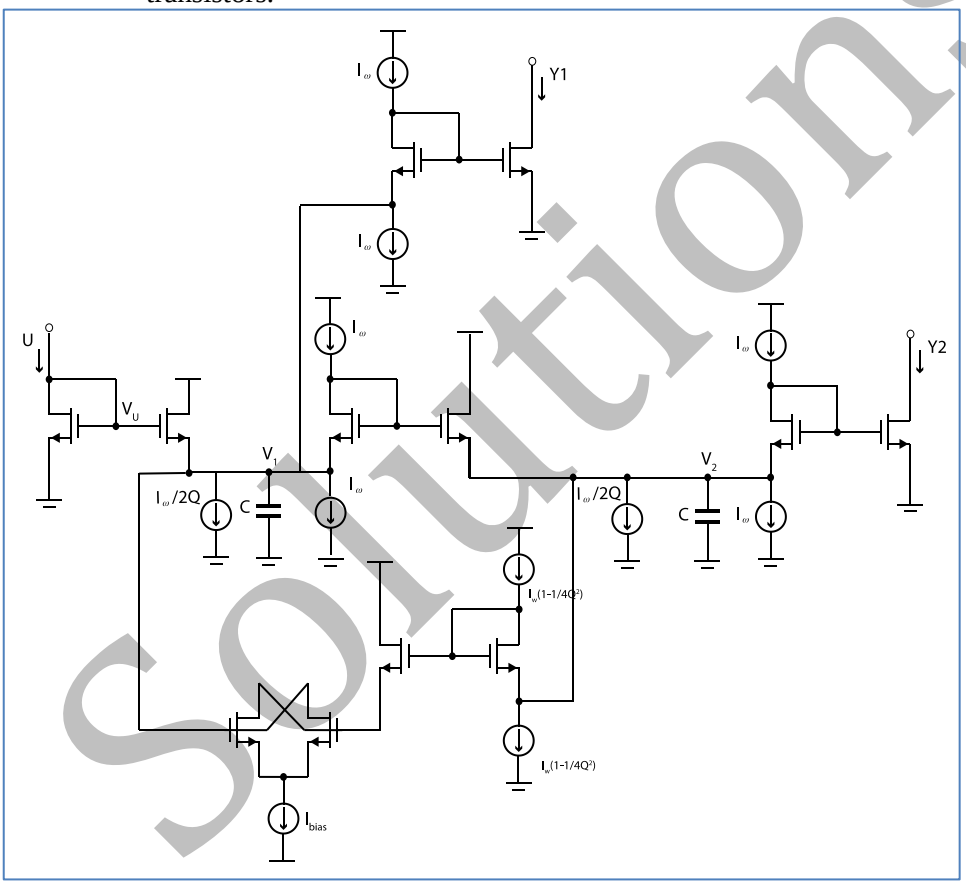
$$I_1 = I_2 = I_\omega$$
$$I_U = I_0$$

Final Equations:

$$C\dot{V}_1 = -\left(\frac{I_\omega}{2Q}\right) - I_\omega\left(1 - \frac{1}{4Q^2}\right)\exp\left(\frac{V_2 - V_1}{nU_t}\right) + I_o \exp\left(\frac{V_u - V_1}{nU_t}\right)$$
$$C\dot{V}_2 = -\left(\frac{I_\omega}{2Q}\right) + I_\omega \exp\left(\frac{V_1 - V_2}{nU_t}\right)$$
$$Y_1 = I_\omega \exp\left(\frac{V_1}{nU_t}\right)$$
$$Y_2 = I_\omega \exp\left(\frac{V_2}{nU_t}\right)$$

[5]

iii. With these log-domain design equations, sketch a schematic of the final log domain topology which realises the two outputs Y_1 and Y_2 using weak inversion MOS transistors.



[5]

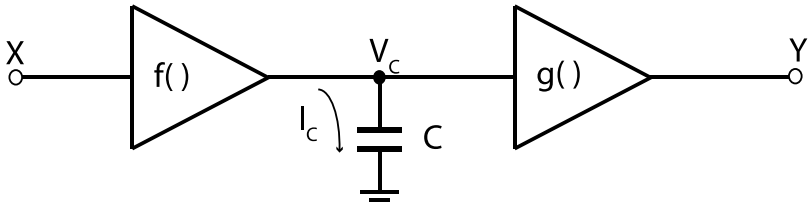


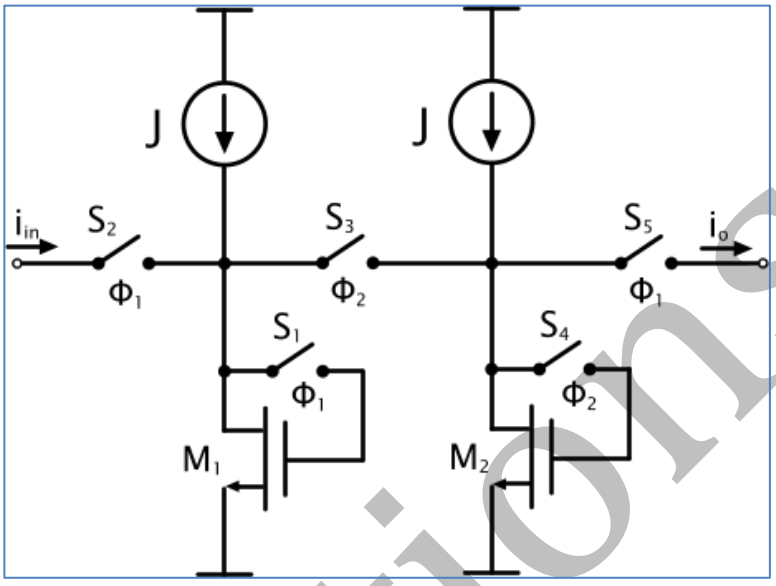
Figure 3.1

Solutions

4.

a) Figure 4.1 depicts a switched current memory cell. Using this memory cell, design a unit delay cell proving a z^{-1} function and derive the equation for its operation.

The delay cell consists of alternately clocked memory cells.



The operation is described as follows.
On phase Φ_1 of clock period (n-1):

$$I_1(n-1) = I_{in}(n-1) + J$$

On phase Φ_2 of clock period (n-1):

$$I_2(n-1) = J - I_{in}(n-1)$$

On phase Φ_1 of clock period (n):

$$I_o(n) = J - I_2(n-1) = I_{in}(n-1)$$

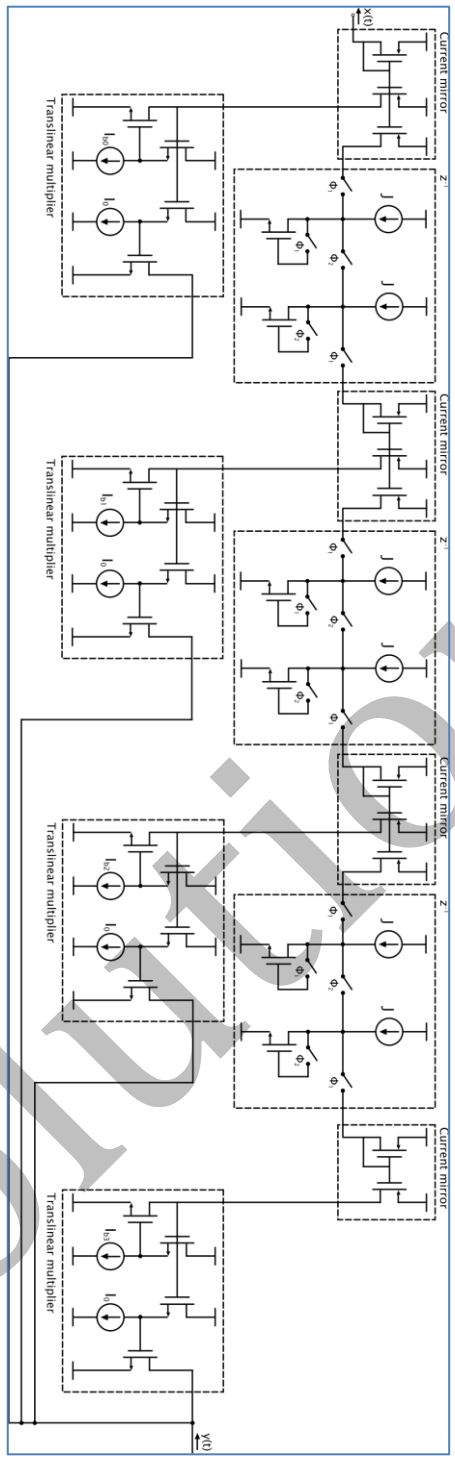
Therefore, the current is delayed and the transfer function is given by

$$\frac{I_o}{I_{in}}(z) = z^{-1}$$

b) Figure 4.2 depicts a FIR filter architecture. Using your unit delay cell from part 4.b sketch a full circuit schematic that implements this function. You may assume the current is unidirectional and use only pMOS current mirrors to copy the current. The coefficients b0-3 must be tunable. Hint : they can be represented by current ratios I_{b0}/I_0, I_{b1}/I_0, I_{b2}/I_0 and I_{b3}/I_0, where I_0 is a MOSFET current in weak inversion.

[8]

Solutions



Solutions

c) Figure 4.3 shows a schematic of a switched current integrator and its associated timing diagram. Derive the transfer function of the switched current integrator and find its time constant. You may assume that the switches are ideal and that the transistors sizing is as follows: $\left(\frac{W}{L}\right)_{M_1} = \left(\frac{W}{L}\right)_{M_2}$, $\left(\frac{W}{L}\right)_{M_4} = b_4\left(\frac{W}{L}\right)_{M_1}$ and $\left(\frac{W}{L}\right)_{M_3} = b_3\left(\frac{W}{L}\right)_{M_1}$.

[8]

Clock period (n-1), phase Φ_2

The current through M_1 is given by

$$I_1(n-1) = 2I - I_2(n-1)$$

where the gate of M_2 is floating and therefore its current mirrors the one of M_4

$$b_4 I_2(n-1) = I_4(n-1)$$
$$I_2(n-1) = I + \frac{i_i(n-1)}{b_4}$$

Hence

$$I_1(n-1) = I - \frac{i_i(n-1)}{b_4}$$

Following the same reasoning for M_3,

$$b_3 I_2(n-1) = I_3(n-1)$$
$$I_1(n-1) = I + \frac{i_0(n-1)}{b_3}$$

Clock period (n), phase Φ_1

First we express the value of $i_x(n)$ with respect to $i_o(n)$ by considering that the currents through M4 and M5 are proportional.

$$\frac{I_4(n)}{b_4} = \frac{I_3(n)}{b_3}$$
$$\Rightarrow i_x(n) = \frac{b_4}{b_3} i_o(n)$$

The current through M_1 has been stored from phase Φ_2

$$I_1(n) = I_1(n-1)$$

KCL on the left part of the circuit:

$$I_1(n) + I_2(n) = 2I + i_{in}(n) + i_x(n)$$
$$I + \frac{i_o(n-1)}{b_3} + I - \frac{i_o(n)}{b_3} = 2I + i_{in}(n) + \frac{b_4}{b_3} i_o(n)$$
$$(1 + b_4) i_o(n) - i_o(n-1) = -b_3 i_{in}(n)$$

The transfer function is then

$$\frac{i_o(z)}{i_i(z)} = -\frac{b_3}{(1 + b_4) - z^{-1}}$$

The circuit is an inverting lossy integrator. To find the time constant, we substitute

$$z = e^{j\omega t} \Rightarrow z^{-1} = e^{-j\omega t} \approx 1 - j\omega t \;\; when \;\; \omega T \ll 1$$

Therefore

$$\frac{i_o(z)}{i_i(z)} = -\frac{b_3/b_4}{1 + j\omega\left(\frac{T}{b_4}\right)}$$

And the time constant is T/b_4.

Solutions

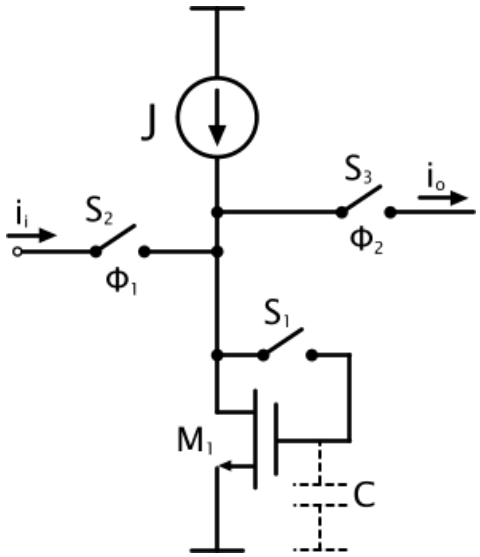


Figure 4.1: Switched current memory cell

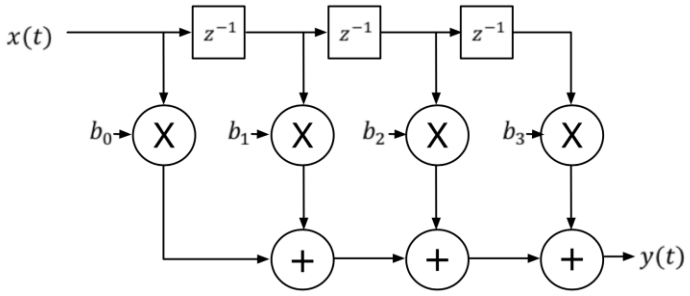


Figure 4.2: FIR filter architecture

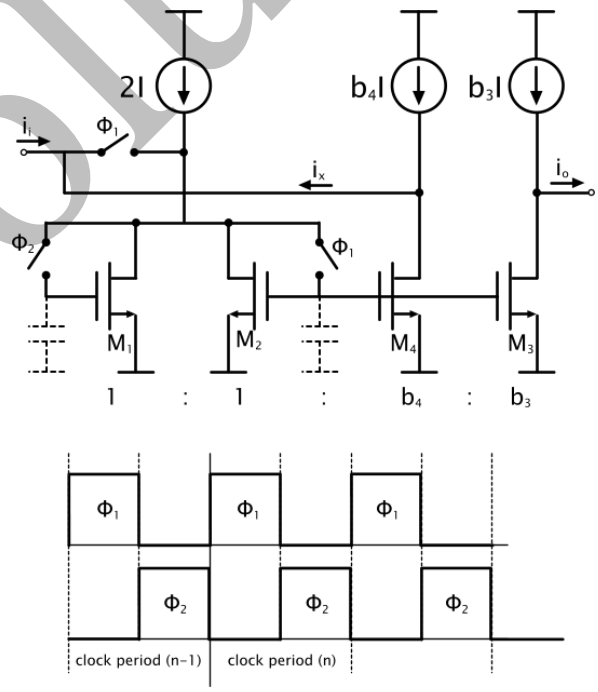


Figure 4.3: Switched current integrator

Solutions

5. a) Figure 5.1 shows a NMOS sampling switch to be used in a switched capacitor circuit. The parameters of M1 are $(W/L)_1 = 30\mu m/1\mu m$, $C_{ox} = 17$ fF/µm^2, $V_{th} = 0.3$ V, CK = $V_{DD} = 1.8$ V and $C_H = 0.1$ pf. The total charge in the inversion layer, when the switch is on, is given by:

$$Q_{ch} = WLC_{ox}(V_{DD} - V_{in} - V_{th})$$

i) Explain the errors caused due to charge injection and clock feedthrough. Derive equations that show how the output voltage will change due to these errors, assuming the charge injection is split equally between the source and the drain.

[4]

1. Charge injection

When a MOSFET is on, there is a charge in the inversion layer:

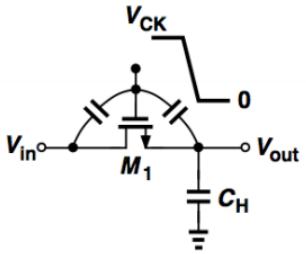
$$Q_{ch} = WLC_{ox}(V_{DD} - V_{in} - V_{th})$$

When the MOSFET turns off the charge must exit through the source and drain terminals. Assuming equal splitting, the error associated with the charge injected onto a capacitor C is

$$\Delta V = \frac{WLC_{ox}(V_{DD} - V_{in} - V_{th})}{2C_H}$$

2. Clock feedthrough

MOS switch couples the clock transitions to the sampling capacitor through its GD and GS overlap capacitances.



$$\Delta V = V_{CK}\frac{WC_{ov}}{WC_{ov} + C_H},$$

ii) Using the output voltage equations derived in 5i), calculate the total change in output voltage due to charge injection and the total change due to clock feed-through. Based on your result, specify which is more significant. You may assume the overlap capacitance $C_{ov} = 0.08$ fF/m and an input voltage $V_{in} = 1$ V.

[3]

Charge injection:

$$\Delta V = \frac{WLC_{ox}(V_{DD} - V_{in} - V_{th})}{2C_H} = \frac{30 \times 1 \times 17f \times (1.8 - 1 - 0.3)}{2 \times 0.1p} = 1.275\ V$$

Clock feedthrough:

$$\Delta V = V_{CK}\frac{WC_{ov}}{WC_{ov} + C_H} = 43.2\ nV$$

Charge injection is therefore more significant.

iii) Show that the output voltage contribution due to charge injection results in a non-unity gain error and constant offset. Plot a graph indicating the value of the offset and the slope.

[3]

The output voltage is expressed as

$$V_{out} = V_{in} - \frac{WLC_{ox}(V_{DD} - V_{in} - V_{th})}{2C_H}$$

Solutions

$$= V_{in}\left(1 + \frac{WLC_{ox}}{2C_H}\right) - \frac{WLC_{ox}}{2C_H}(V_{DD} - V_{th})$$

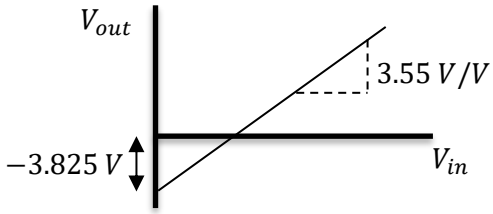
In this equation,

$$\frac{WLC_{ox}}{2C_H} = 2.55\ V$$

Hence V_{out} expressed in V is given by

$$V_{out} = 3.55\ V_{in} - 3.825$$

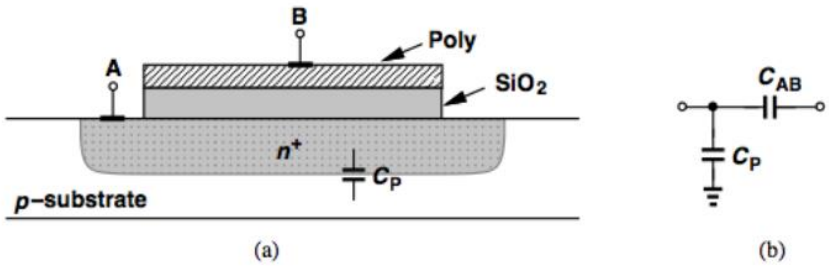
The slope is 3.55 V/V and the offset is -3.825 V



b) Figure 5.2 shows a circuit diagram of a differential switched capacitor non-inverting amplifier.

i) Explain the necessity of bottom plate sampling in this design and draw a schematic of the amplifier indicating which sides of the capacitors are the bottom plates. [5]

Capacitors implemented in CMOS have a top polysilicon layer and a bottom plate which is a heavily doped n+ region. Therefore, the bottom plate has a parasitic junction capacitance Cp to the substrate which is usually about 10-20% of the oxide capacitance of the MOSFET.

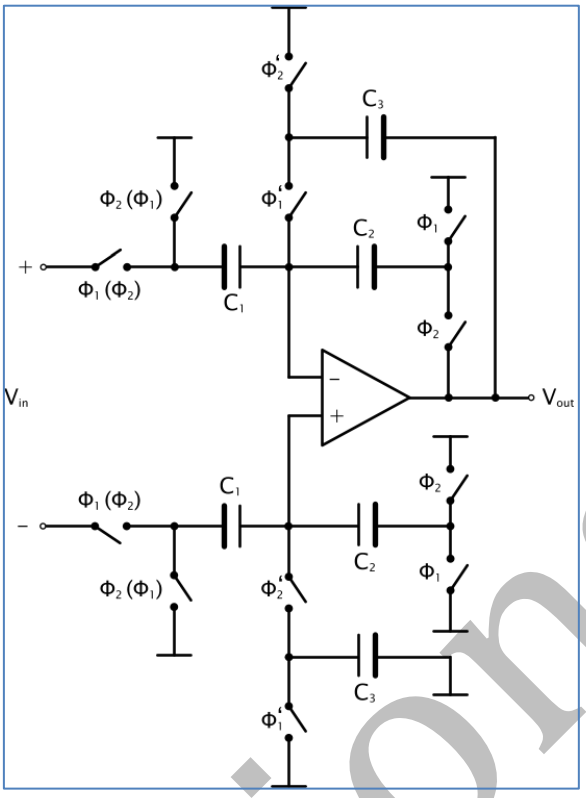


(a) Monolithic capacitor structure, (b) circuit model of (a) including parasitic capacitance to the substrate.

This capacitance is variable and should not be connected to the input of the op amp, as the input capacitance affects speed and precision and needs to be well defined. Therefore we always connect the top plate to the input of the op amp (high impedance node), and the bottom plate to a low impedance node which in this design is the input. This minimises the effect of the parasitic capacitance and avoids injection of substrate noise.

The schematic is drawn hereunder with the thick lines in each capacitor representing the bottom plate.

Solutions



ii) Derive the transfer function of the Figure 5.2 and also derive an equation to show the change in output voltage during phases Φ_1 and Φ_2.

[5]

Let us start with the top part of the circuit.

Phase Φ_1 :

$$Q_1 = C_1 \times (V_{in+} - V_{x-})$$
$$Q_2 = C_2 \times (V_{x-} - 0)$$
$$Q_3 = C_3 \times (V_{x-} - V_{out}^{\Phi_1})$$

Phase Φ_2 :

$$Q_1 = C_1 \times (0 - V_{x-})$$
$$Q_2 = C_2 \times (V_{x-} - V_{out}^{\Phi_2})$$
$$Q_3 = C_3 \times (0 - V_{out}^{\Phi_2})$$

Charge conservation at node V_{x-} in phase Φ_2:

$$Q_2^{\Phi_2} - Q_2^{\Phi_1} = Q_1^{\Phi_2} - Q_1^{\Phi_1}$$
$$C_2\left(V_{x-} - V_{out}^{\Phi_2}\right) - C_2 V_{x-} = -C_1 V_{x-} - C_1(V_{in+} - V_{x-})$$
$$C_2 V_{out}^{\Phi_2} = C_1 V_{in+}$$

Now the bottom part of the circuit.

Phase Φ_1 :

$$Q_1 = C_1 \times (V_{in-} - V_{x+})$$
$$Q_2 = C_2 \times (V_{x+} - 0)$$
$$Q_3 = C_3 \times (0)$$

Phase Φ_2 :

Solutions

$$Q_1 = C_1 \times (0 - V_{x+})$$
$$Q_2 = C_2 \times (V_{x+} - 0)$$
$$Q_3 = C_3 \times (V_{x+})$$

Charge conservation at node V_{x+} in phase Φ_1:

$$Q_2^{\Phi_2} - Q_2^{\Phi_1} = Q_1^{\Phi_2} - Q_1^{\Phi_1}$$
$$C_2 V_{x+} - C_2 V_{x+} = -C_1 V_{x+} - C_1 (V_{in-} - V_{x+})$$
$$V_{in-} = 0$$

which means that input contribution is given by the top circuit and the differential architecture can be considered like its single-ended equivalent. The transfer function is:

$$V_{out}^{\Phi_2} = \frac{C_1}{C_2} V_{in}$$

Following charge conservation on C_3, the difference between output voltages in phases Φ_1 and Φ_2 is given by

$$\Delta V = V_{out}^{\Phi_1} - V_{out}^{\Phi_2} = V_{x-} = V_{os} - \frac{V_{out}}{A}$$

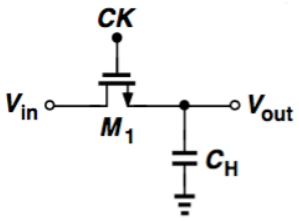


Figure 5.1

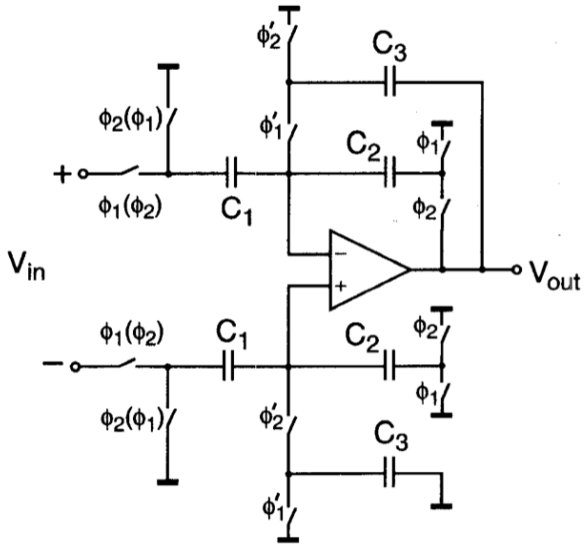


Figure 5.2

Solutions

6. Figure 6.1 shows a measurement system that digitises the product of two different signals. The architecture uses an operational amplifier, a current multiplying circuit and an analogue-to-digital converter to encode the detected voltage signal. The detected signals are first converted to currents through a voltage-to-current converter (V/I), then multiplied in current mode and finally converted to a spike train through an integrate-and-fire neuron (I/F). The 8 bit counter is used to count the number of spikes in a fixed period T_{reset}. The whole system works off a 1 V supply.

a) Describe a technique which allows detection of very small voltage signals which exist below the 1/f noise floor of the operational amplifier. Include diagrams showing the operation of this technique and how the frequency spectrum of the signal and noise is shaped in each stage.

[5]

We can use the chopper stabilization technique: We up-modulate our input signal using square wave modulation to a frequency which is above 1/f noise spectrum of our amplifier. We then amplify and then down modulate at the output with the same square wave signal. At the output however noise and offset get up-modulated which can then be removed using a low-pass filter.

[2]

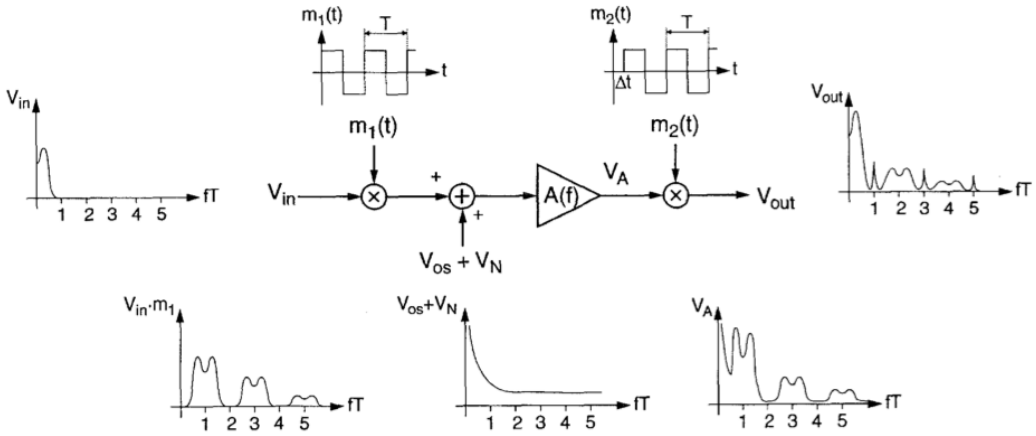


Fig. 9. The chopper amplification principle [19].

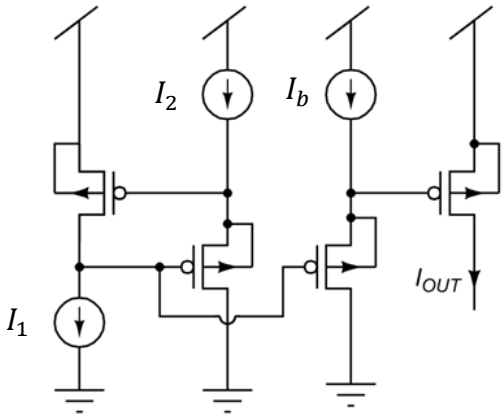
[3]

b) Sketch a circuit for multiplying two currents. State the region of operation for the MOSFETs and derive the relationship between the output and the input. Identify a way to provide a gain to multiplication.

[4]

The circuit is a translinear circuit with all MOSFETs operating in weak inversion.

SOLNS



Following the translinear principle,

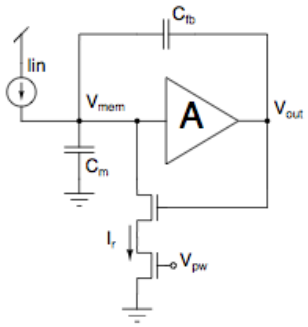
$$I_{out} = \frac{I_1 I_2}{I_b}$$

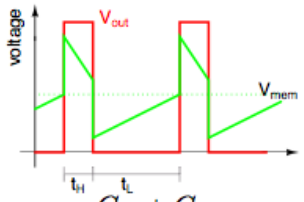
Hence on this circuit, the output current is the multiplication of I_1
The circuit can provide an amplification factor by tuning the value of I_b.

c) Sketch an integrate-and-fire circuit which can be used in this design and describe its operation. Derive equations for the pulse width and frequency.

[5]

We can use an axon-hillock integrate and fire neuron. Current is used to charge the capacitor which resets the voltage after it passes the threshold of the amplifier causing a spike.





$$t_L = \frac{C_{fb} + C_m}{I_{in}} \Delta V_{mem} = \frac{C_{fb}}{I_{in}} V_{dd}$$

$$t_H = \frac{C_{fb} + C_m}{I_r - I_{in}} \Delta V_{mem} = \frac{C_{fb}}{I_r - I_{in}} V_{dd}$$

Frequency $\propto I_{in}$

Pulse width $\propto 1/I_r$ for $I_r \gg I_{in}$

[2 points for circuit and explanation, 1point for graph, 2 points for derivations]

[Total 5 points]

d) i. Given that the maximum output is limited to $I_{out} = 100$ nA, chose a suitable capacitor value for your integrate-and-fire circuit such that the spike frequency is always less than 5 kHz. You may assume the refractory current is $I_r = 1$ nA.

V_{dd}=1V, I_{in}=100nA, I_r=1nA
T_{total}= t_n + t_l = 1/5KHz
C_{fd}. V_{dd}/I_{in} + C_{fd}. $V_{dd}/(I_r - I_{in})$ = 1/5KHz

C_{fd}= 10pf

[4]

ii. Calculate the maximum allowable reset time T_{rst} such that the counter never overflows for this current. How many bits are required when integrating over a long time constant $T_{rst} = 1\ s$?

[2]

8 bit counter means 2^8 counts therefore Maximum $T_{reset} = 2^8/5x10^3 = 51.2$ msec. [1]
For T_{reset}=1 second we need n=12.28= 13bit counter. [1]

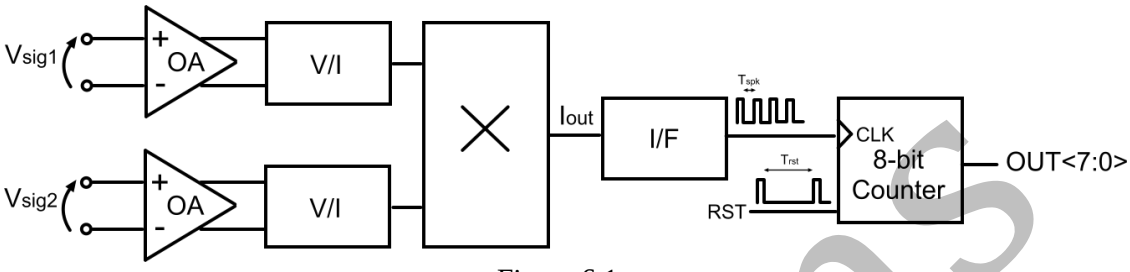


Figure 6.1

Solutions